

Class 13: Transcriptomics and RNA-seq Analysis

Julianne Lee (PID: A17979401)

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Bioconductor setup

```
library(BiocManager)
```

```
Bioconductor version '3.20' is out-of-date; the current release version '3.23'
is available with R version '4.6'; see https://bioconductor.org/install
```

```
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,
table, tapply, union, unique, unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: GenomeInfoDb

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

Import countData and colData

```
counts <- read.csv("airway_scaledcounts.csv", row.names = 1)
metadata <- read.csv("airway_metadata.csv")

head(counts)
```

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG00000000003	723	486	904	445	1170
ENSG00000000005	0	0	0	0	0
ENSG000000000419	467	523	616	371	582
ENSG000000000457	347	258	364	237	318
ENSG000000000460	96	81	73	66	118
ENSG000000000938	0	0	1	0	2

	SRR1039517	SRR1039520	SRR1039521
ENSG00000000003	1097	806	604
ENSG00000000005	0	0	0
ENSG000000000419	781	417	509
ENSG000000000457	447	330	324
ENSG000000000460	94	102	74
ENSG000000000938	0	0	0

```
head(metadata)
```

	id	dex	celltype	geo_id
1	SRR1039508	control	N61311	GSM1275862
2	SRR1039509	treated	N61311	GSM1275863
3	SRR1039512	control	N052611	GSM1275866
4	SRR1039513	treated	N052611	GSM1275867
5	SRR1039516	control	N080611	GSM1275870
6	SRR1039517	treated	N080611	GSM1275871

Q1. How many genes are in this dataset?

```
nrow(counts)
```

```
[1] 38694
```

Q2. How many 'control' cell lines do we have?

```
sum(metadata$dex == "control")
```

```
[1] 4
```

Toy differential gene expression

```
control <- metadata[metadata[, "dex"]=="control",]  
control.counts <- counts[ ,control$id]  
control.mean <- rowSums( control.counts )/4  
head(control.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460  
          900.75          0.00          520.50          339.75          97.25  
ENSG0000000000938  
          0.75
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following object is masked from 'package:Biobase':

combine

The following object is masked from 'package:matrixStats':

count

The following objects are masked from 'package:GenomicRanges':

intersect, setdiff, union

The following object is masked from 'package:GenomeInfoDb':

intersect

The following objects are masked from 'package:IRanges':

collapse, desc, intersect, setdiff, slice, union

The following objects are masked from 'package:S4Vectors':

first, intersect, rename, setdiff, setequal, union

The following objects are masked from 'package:BiocGenerics':

combine, intersect, setdiff, union

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
control <- metadata %>% filter(dex=="control")
control.counts <- counts %>% select(control$id)
control.mean <- rowSums(control.counts)/4
head(control.mean)
```

```
ENSG00000000003 ENSG00000000005 ENSG00000000419 ENSG00000000457 ENSG00000000460
      900.75           0.00           520.50           339.75           97.25
ENSG000000000938
      0.75
```

Q3. How would you make the above code in either approach more robust? Is there a function that could help here?

```
control <- metadata[metadata[, "dex"]=="control",]
control.counts <- counts[, control$id]
control.mean <- rowMeans(control.counts)
head(control.mean)
```

```
ENSG00000000003 ENSG00000000005 ENSG00000000419 ENSG00000000457 ENSG00000000460
      900.75           0.00           520.50           339.75           97.25
ENSG000000000938
      0.75
```

Q4. Follow the same procedure for the treated samples (i.e. calculate the mean per gene across drug treated samples and assign to a labeled vector called treated.mean)

```
treated <- metadata[metadata$dex == "treated", ]
treated.counts <- counts[, treated$id]
treated.mean <- rowMeans(treated.counts)

head(treated.mean)
```

```
ENSG000000000003  ENSG000000000005  ENSG000000000419  ENSG000000000457  ENSG000000000460
                658.00                0.00                546.00                316.50                78.75
ENSG000000000938
                0.00
```

```
meancounts <- data.frame(control.mean, treated.mean)
```

Q5 (a). Create a scatter plot showing the mean of the treated samples against the mean of the control samples. Your plot should look something like the following.

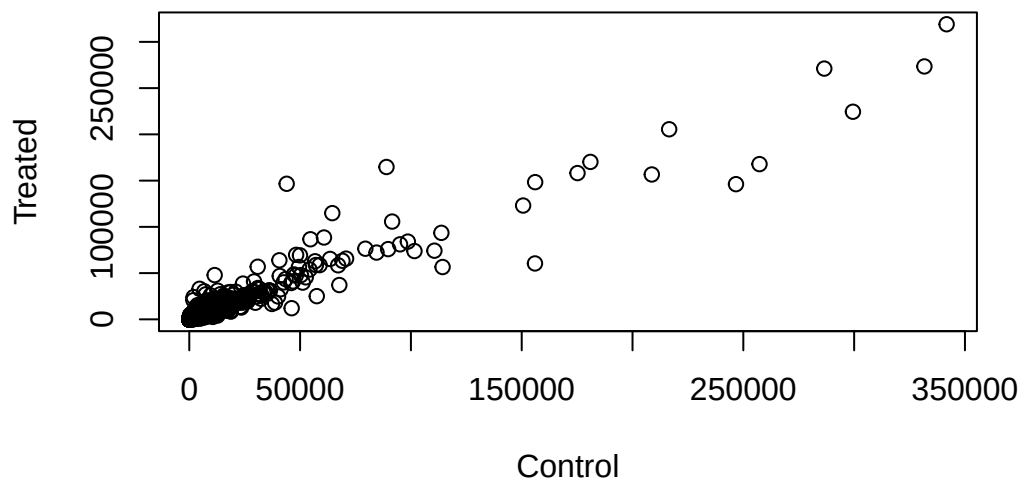
```
meancounts <- data.frame(control.mean, treated.mean)
head(meancounts)
```

	control.mean	treated.mean
ENSG000000000003	900.75	658.00
ENSG000000000005	0.00	0.00
ENSG000000000419	520.50	546.00
ENSG000000000457	339.75	316.50
ENSG000000000460	97.25	78.75
ENSG000000000938	0.75	0.00

```
colSums(meancounts)
```

```
control.mean  treated.mean
      23005324      22196524
```

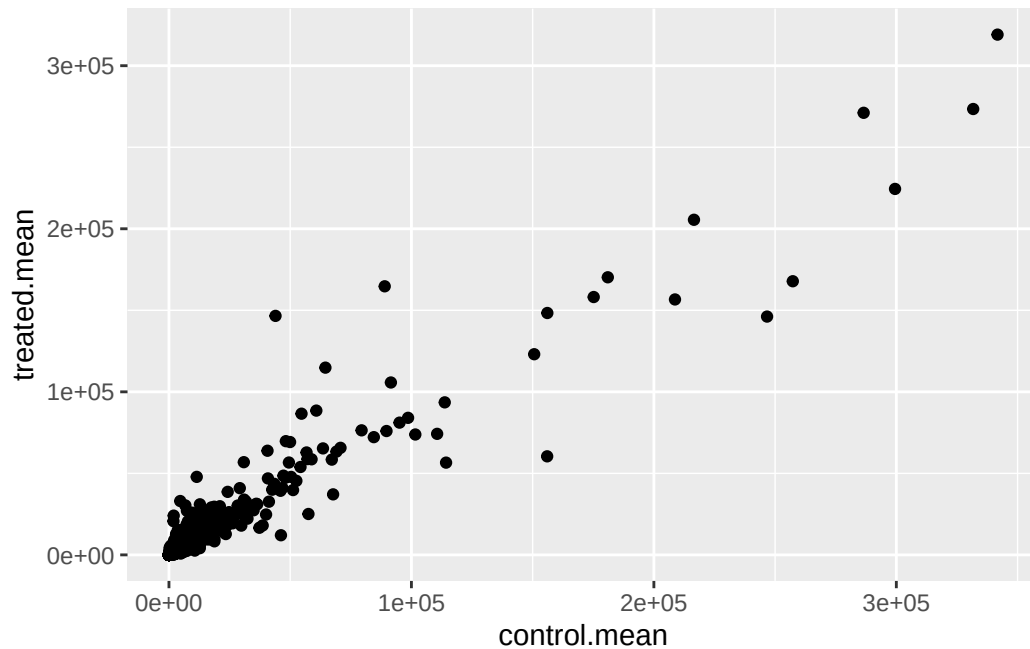
```
plot(meancounts[,1], meancounts[,2],
     xlab = "Control",
     ylab = "Treated")
```



Q5 (b). You could also use the ggplot2 package to make this figure producing the plot below. What `geom_?()` function would you use for this plot?

```
library(ggplot2)

ggplot(meancounts, aes(x = control.mean, y = treated.mean)) +
  geom_point()
```

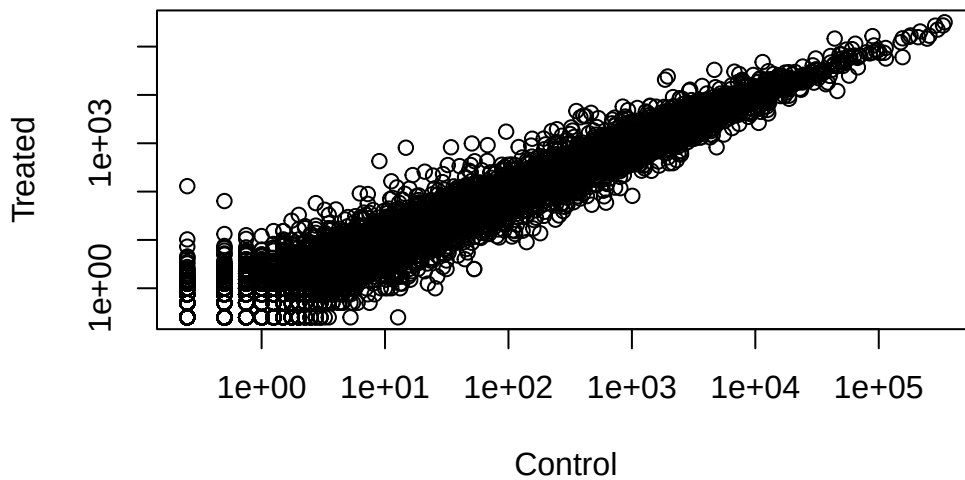



Q6. Try plotting both axes on a log scale. What is the argument to `plot()` that allows you to do this?

```
plot(meancounts$control.mean, meancounts$treated.mean,
     log = "xy",
     xlab = "Control",
     ylab = "Treated")
```

Warning in `xy.coords(x, y, xlabel, ylabel, log)`: 15032 x values ≤ 0 omitted from logarithmic plot

Warning in `xy.coords(x, y, xlabel, ylabel, log)`: 15281 y values ≤ 0 omitted from logarithmic plot



```
meancounts$log2fc <- log2(meancounts[, "treated.mean"] / meancounts[, "control.mean"])
head(meancounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000005	0.00	0.00	NaN
ENSG0000000000419	520.50	546.00	0.06900279
ENSG0000000000457	339.75	316.50	-0.10226805
ENSG0000000000460	97.25	78.75	-0.30441833
ENSG0000000000938	0.75	0.00	-Inf

```
zero.vals <- which(meancounts[, 1:2] == 0, arr.ind = TRUE)

to.rm <- unique(zero.vals[, 1])
mycounts <- meancounts[-to.rm, ]
head(mycounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG0000000000419	520.50	546.00	0.06900279
ENSG0000000000457	339.75	316.50	-0.10226805
ENSG0000000000460	97.25	78.75	-0.30441833

ENSG00000000971	5219.00	6687.50	0.35769358
ENSG00000001036	2327.00	1785.75	-0.38194109

Q7. What is the purpose of the `arr.ind` argument in the `which()` function call above? Why would we then take the first column of the output and need to call the `unique()` function?

```
zero.vals <- which(meancounts[, 1:2] == 0, arr.ind = TRUE)
#zero.vals
```

`arr.ind` makes `which()` to return the row and column positions of the values that match the condition. We take first column because the first column contains the gene into to be removed.

Q8. Using the `up.ind` vector above can you determine how many up regulated genes we have at the greater than 2 fc level?

```
up.ind <- mycounts$log2fc > 2
sum(up.ind)
```

[1] 250

Q9. Using the `down.ind` vector above can you determine how many down regulated genes we have at the greater than 2 fc level?

```
down.ind <- mycounts$log2fc < -2
sum(down.ind)
```

[1] 367

Q10. Do you trust these results? Why or why not?

No, since this uses log2 fold change. The result can be unreliable if the read counts are low or highly variable. Also, this is not performing statistical analysis.

Setting up for DESeq

```
citation("DESeq2")
```

To cite package 'DESeq2' in publications use:

Love, M.I., Huber, W., Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2 Genome Biology 15(12):550 (2014)

A BibTeX entry for LaTeX users is

```
@Article{,  
  title = {Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2},  
  author = {Michael I. Love and Wolfgang Huber and Simon Anders},  
  year = {2014},  
  journal = {Genome Biology},  
  doi = {10.1186/s13059-014-0550-8},  
  volume = {15},  
  issue = {12},  
  pages = {550},  
}
```

```
dds <- DESeqDataSetFromMatrix(countData=counts,  
                              colData=metadata,  
                              design=~dex)
```

converting counts to integer mode

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds
```

```
class: DESeqDataSet  
dim: 38694 8  
metadata(1): version  
assays(1): counts  
rownames(38694): ENSG000000000003 ENSG000000000005 ... ENSG00000283120
```

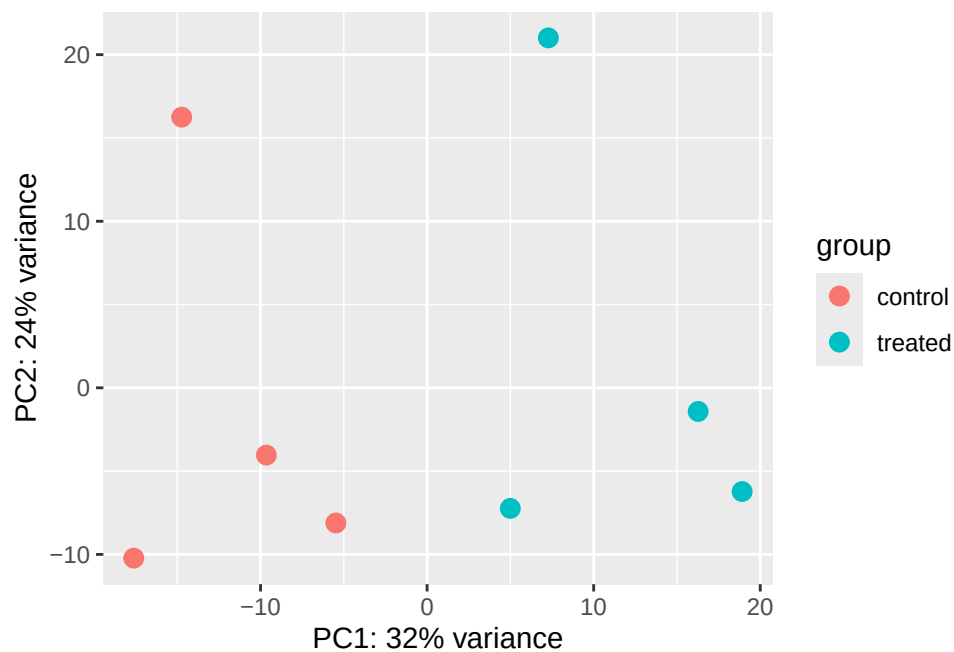
```
ENSG00000283123
rowData names(0):
colnames(8): SRR1039508 SRR1039509 ... SRR1039520 SRR1039521
colData names(4): id dex celltype geo_id
```

Principal Component Analysis (PCA)

```
vsd <- vst(dds, blind = FALSE)
plotPCA(vsd, intgroup = c("dex"))
```

using ntop=500 top features by variance

Warning: `aes\_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the DESeq2 package.
Please report the issue to the authors.



```
pcaData <- plotPCA(vsd, intgroup=c("dex"), returnData=TRUE)
```

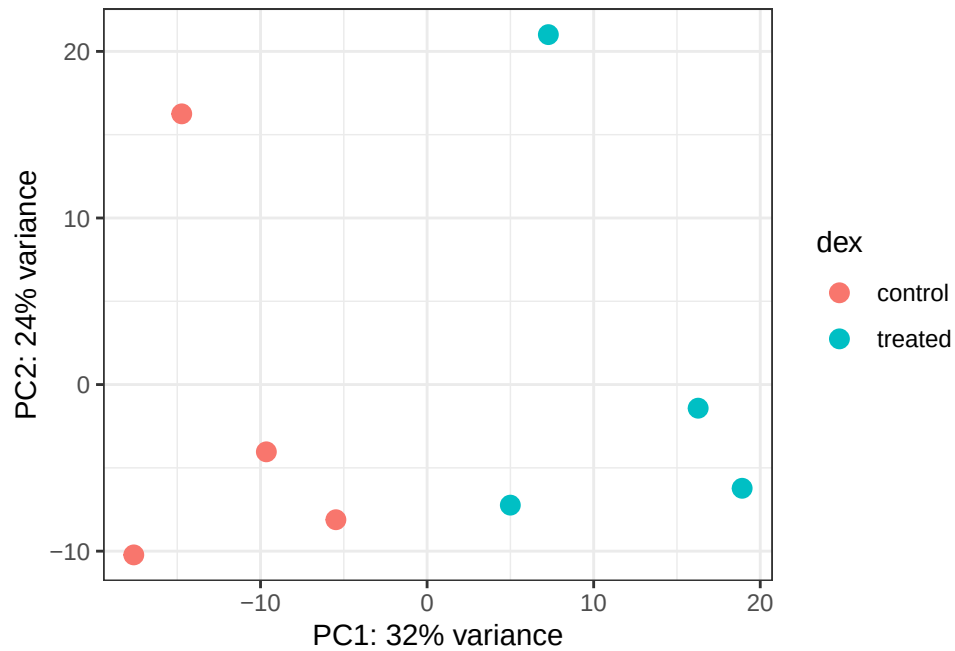
using ntop=500 top features by variance

```
head(pcaData)
```

	PC1	PC2	group	dex	name
SRR1039508	-17.607922	-10.225252	control	control	SRR1039508
SRR1039509	4.996738	-7.238117	treated	treated	SRR1039509
SRR1039512	-5.474456	-8.113993	control	control	SRR1039512
SRR1039513	18.912974	-6.226041	treated	treated	SRR1039513
SRR1039516	-14.729173	16.252000	control	control	SRR1039516
SRR1039517	7.279863	21.008034	treated	treated	SRR1039517

```
percentVar <- round(100 * attr(pcaData, "percentVar"))
```

```
ggplot(pcaData) +  
  aes(x = PC1, y = PC2, color = dex) +  
  geom_point(size = 3) +  
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +  
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +  
  coord_fixed() +  
  theme_bw()
```



DESeq analysis

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
res <- results(dds)  
res
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 38694 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.1942	-0.3507030	0.168246	-2.084470	0.0371175
ENSG000000000005	0.0000	NA	NA	NA	NA
ENSG000000000419	520.1342	0.2061078	0.101059	2.039475	0.0414026
ENSG000000000457	322.6648	0.0245269	0.145145	0.168982	0.8658106
ENSG000000000460	87.6826	-0.1471420	0.257007	-0.572521	0.5669691
...	...	...	...	...	...
ENSG00000283115	0.000000	NA	NA	NA	NA
ENSG00000283116	0.000000	NA	NA	NA	NA
ENSG00000283119	0.000000	NA	NA	NA	NA
ENSG00000283120	0.974916	-0.668258	1.69456	-0.394354	0.693319
ENSG00000283123	0.000000	NA	NA	NA	NA
	padj				
	<numeric>				
ENSG000000000003	0.163035				
ENSG000000000005	NA				
ENSG000000000419	0.176032				
ENSG000000000457	0.961694				
ENSG000000000460	0.815849				
...	...				
ENSG00000283115	NA				
ENSG00000283116	NA				
ENSG00000283119	NA				
ENSG00000283120	NA				
ENSG00000283123	NA				

```
summary(res)
```

out of 25258 with nonzero total read count

adjusted p-value < 0.1

LFC > 0 (up) : 1563, 6.2%

LFC < 0 (down) : 1188, 4.7%

outliers [1] : 142, 0.56%

low counts [2] : 9971, 39%

(mean count < 10)

[1] see 'cooksCutoff' argument of ?results

[2] see 'independentFiltering' argument of ?results


```
res05 <- results(dds, alpha=0.05)
summary(res05)
```

```
out of 25258 with nonzero total read count
adjusted p-value < 0.05
LFC > 0 (up)      : 1236, 4.9%
LFC < 0 (down)    : 933, 3.7%
outliers [1]      : 142, 0.56%
low counts [2]    : 9033, 36%
(mean count < 6)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Adding annotation data

```
#BiocManager::install("AnnotationDbi")
#BiocManager::install("org.Hs.eg.db")
```

```
library("AnnotationDbi")
```

Attaching package: 'AnnotationDbi'

The following object is masked from 'package:dplyr':

select

```
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

```
[1] "ACCNUM"      "ALIAS"      "ENSEMBL"    "ENSEMBLPROT" "ENSEMBLTRANS"
[6] "ENTREZID"    "ENZYME"     "EVIDENCE"   "EVIDENCEALL" "GENENAME"
[11] "GENETYPE"    "GO"         "GOALL"      "IPI"         "MAP"
[16] "OMIM"        "ONTOLOGY"   "ONTOLOGYALL" "PATH"        "PFAM"
[21] "PMID"        "PROSITE"    "REFSEQ"     "SYMBOL"      "UCSCKG"
[26] "UNIPROT"
```

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="SYMBOL",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 7 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.3507030	0.168246	-2.084470	0.0371175
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG0000000000419	520.134160	0.2061078	0.101059	2.039475	0.0414026
ENSG0000000000457	322.664844	0.0245269	0.145145	0.168982	0.8658106
ENSG0000000000460	87.682625	-0.1471420	0.257007	-0.572521	0.5669691
ENSG0000000000938	0.319167	-1.7322890	3.493601	-0.495846	0.6200029

	padj	symbol
	<numeric>	<character>
ENSG0000000000003	0.163035	TSPAN6
ENSG0000000000005	NA	TNMD
ENSG00000000000419	0.176032	DPM1
ENSG00000000000457	0.961694	SCYL3
ENSG00000000000460	0.815849	FIRRM
ENSG00000000000938	NA	FGR

Q11. Run the mapIds() function two more times to add the Entrez ID and UniProt accession and GENENAME as new columns called *resentrez*, *resuniprot* and *res\$genename*.

```
res$entrez <- mapIds(org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "ENTREZID",
  multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$uniprot <- mapIds(org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "UNIPROT",
  multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$genename <- mapIds(org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "GENENAME",
  multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 10 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.3507030	0.168246	-2.084470	0.0371175
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG000000000419	520.134160	0.2061078	0.101059	2.039475	0.0414026
ENSG000000000457	322.664844	0.0245269	0.145145	0.168982	0.8658106
ENSG000000000460	87.682625	-0.1471420	0.257007	-0.572521	0.5669691
ENSG000000000938	0.319167	-1.7322890	3.493601	-0.495846	0.6200029
	padj	symbol	entrez	uniprot	
	<numeric>	<character>	<character>	<character>	

ENSG000000000003	0.163035	TSPAN6	7105	AOA087WYV6
ENSG000000000005	NA	TNMD	64102	Q9H2S6
ENSG000000000419	0.176032	DPM1	8813	H0Y368
ENSG000000000457	0.961694	SCYL3	57147	X6RHX1
ENSG000000000460	0.815849	FIRRM	55732	A6NFP1
ENSG000000000938	NA	FGR	2268	B7Z6W7

```

      genename
      <character>
ENSG000000000003      tetraspanin 6
ENSG000000000005      tenomodulin
ENSG000000000419 dolichyl-phosphate m..
ENSG000000000457 SCY1 like pseudokina..
ENSG000000000460 FIGNL1 interacting r..
ENSG000000000938 FGR proto-oncogene, ..

```

```
ord <- order( res$padj )
head(res[ord,])
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 10 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000152583	954.771	4.36836	0.2371268	18.4220	8.74490e-76
ENSG00000179094	743.253	2.86389	0.1755693	16.3120	8.10784e-60
ENSG00000116584	2277.913	-1.03470	0.0650984	-15.8944	6.92855e-57
ENSG00000189221	2383.754	3.34154	0.2124058	15.7319	9.14433e-56
ENSG00000120129	3440.704	2.96521	0.2036951	14.5571	5.26424e-48
ENSG00000148175	13493.920	1.42717	0.1003890	14.2164	7.25128e-46

	padj	symbol	entrez	uniprot
	<numeric>	<character>	<character>	<character>
ENSG00000152583	1.32441e-71	SPARCL1	8404	B4E2Z0
ENSG00000179094	6.13966e-56	PER1	5187	A2I2P6
ENSG00000116584	3.49776e-53	ARHGEF2	9181	AOA8Q3SIN5
ENSG00000189221	3.46227e-52	MAOA	4128	B4DF46
ENSG00000120129	1.59454e-44	DUSP1	1843	B4DRR4
ENSG00000148175	1.83034e-42	STOM	2040	F8VSL7

```

      genename
      <character>
ENSG00000152583      SPARC like 1
ENSG00000179094 period circadian reg..
ENSG00000116584 Rho/Rac guanine nucl..

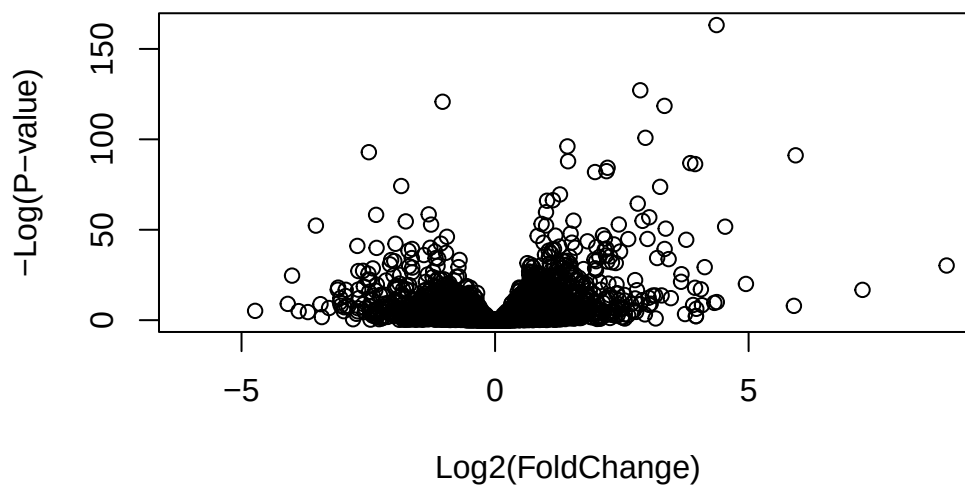
```

```
ENSG00000189221    monoamine oxidase A
ENSG00000120129 dual specificity pho..
ENSG00000148175          stomatin
```

```
write.csv(res[ord,], "deseq_results.csv")
```

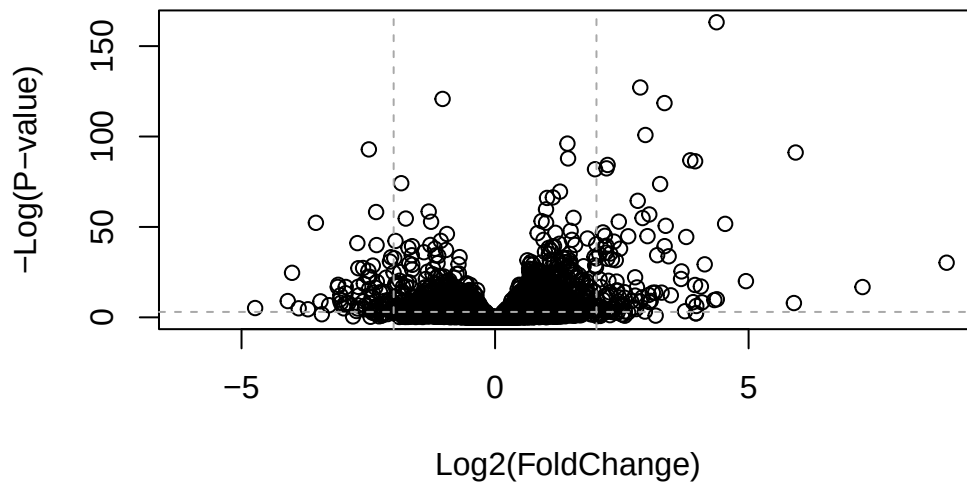
Data Visualization

```
plot( res$log2FoldChange, -log(res$padj),
      xlab="Log2(FoldChange)",
      ylab="-Log(P-value)")
```



```
plot( res$log2FoldChange, -log(res$padj),
      ylab="-Log(P-value)", xlab="Log2(FoldChange)")

abline(v=c(-2,2), col="darkgray", lty=2)
abline(h=-log(0.05), col="darkgray", lty=2)
```

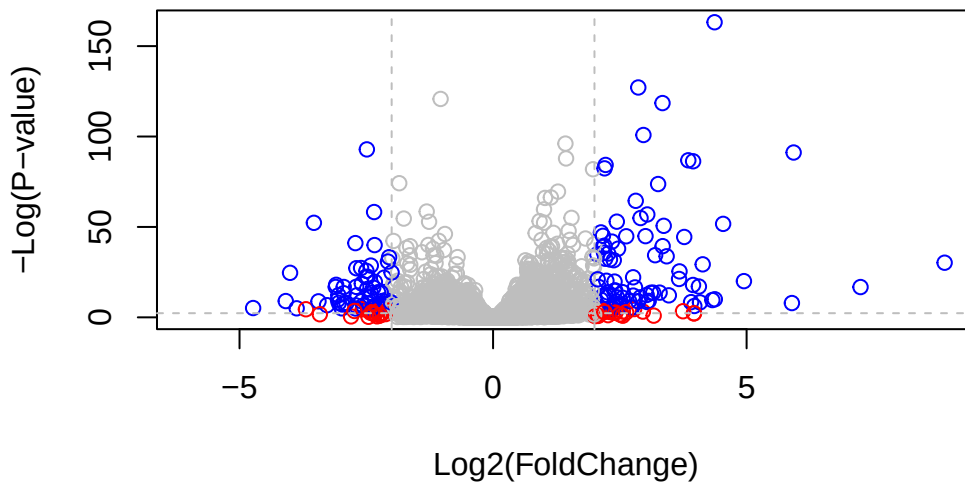


```
mycols <- rep("gray", nrow(res))
mycols[ abs(res$log2FoldChange) > 2 ] <- "red"

inds <- (res$padj < 0.01) & (abs(res$log2FoldChange) > 2 )
mycols[ inds ] <- "blue"

plot( res$log2FoldChange, -log(res$padj),
      col=mycols, ylab="-Log(P-value)", xlab="Log2(FoldChange)" )

abline(v=c(-2,2), col="gray", lty=2)
abline(h=-log(0.1), col="gray", lty=2)
```



```
#BiocManager::install("EnhancedVolcano")
```

```
library(EnhancedVolcano)
```

Loading required package: ggrepel

```
x <- as.data.frame(res)
```

```
EnhancedVolcano(x,
  lab = x$symbol,
  x = 'log2FoldChange',
  y = 'pvalue')
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.
 i The deprecated feature was likely used in the EnhancedVolcano package.
 Please report the issue to the authors.

Warning: The `size` argument of `element\_line()` is deprecated as of ggplot2 3.4.0.
 i Please use the `linewidth` argument instead.
 i The deprecated feature was likely used in the EnhancedVolcano package.
 Please report the issue to the authors.

Volcano plot

EnhancedVolcano

